

HRITHIK NANDI

PERSONAL DATA

DATE OF BIRTH: 18 March 2000
ADDRESS (WORK): Institute for Advancing Intelligence TCG CREST, Kolkata, India
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RESEARCH POSITIONS HELD

Aug. 2022 – Present | **Cryptology and Security Group**, IAI TCG CREST, Kolkata, India
Position: Senior Research Fellow

EDUCATION

Aug. 2022 – Present | **Ph.D. in Computer Science**
Institute for Advancing Intelligence TCG CREST
Ramakrishna Mission Vivekananda Educational and Research Institute
Advisor(s): Dr. Avijit Dutta & Dr. Soumitra Samanta

Aug. 2020 – Jul. 2022 | **M.Sc. in Mathematics**
Ramakrishna Mission Vivekananda Educational and Research Institute

Jul. 2017 – Jun. 2020 | **B.Sc. in Mathematics**
Ramakrishna Mission Vivekananda Centenary College

RESEARCH INTERESTS

- Provable Security
- Design and Analysis of Symmetric Key Cryptographic Schemes
- Post-quantum Security of Symmetric Key Cryptographic Schemes

PUBLICATIONS

1. Nilanjan Datta, Hrithik Nandi, Soumit Pal, Yu Sasaki, Patrick Struck, Maximiliane Weishäupl. **Generic Committing Attacks: Zero-Padded Ascon is Less Secure than Expected.** [[CRYPTO 2026](#), [ePrint](#)]
2. Nilanjan Datta, Jean Paul Degabriele, Avijit Dutta, Vukašin Karadžić, Hrithik Nandi. **Rugged Pseudorandom Permutations with Beyond-Birthday-Bound Security.** [[AsiaCCS 2026](#), [ePrint](#)]
3. Ritam Bhaumik, Nilanjan Datta, Avijit Dutta, Ashwin Jha, Sougata Mandal, Bart Mennink, Hrithik Nandi, Yaobin Shen. **How to Build a Short-Input Random Oracle from Public Random Permutations.** [[EUROCRYPT 2026](#), [ePrint](#)]
4. Nilanjan Datta, Avijit Dutta, Shibam Ghosh, Eik List, Hrithik Nandi. **HCTR+: An Optimally Secure TBC-based Accordion Mode.** [[ToSC 2025\(3\)](#), [ePrint](#)]

5. Nilanjan Datta, Avijit Dutta, Sougata Mandal, Hrithik Nandi. **Sequential Indifferentiability of STH and EDM.**
[CiC 2(2), ePrint]
6. Ritam Bhaumik, Wonseok Choi, Avijit Dutta, Cuauhtemoc Mancillas López, Hrithik Nandi, Yaobin Shen. **Efficient Variants of TNT with BBB Security.**
[ProvSec 2024, ePrint]

PREPRINTS

1. Nilanjan Datta, Avijit Dutta, Sougata Mandal, Hrithik Nandi, Amlan Sinha. **Post-Quantum Security of Keyed Sum of Permutations and Its Siblings.**
[ePrint]
2. Sougata Mandal, Hrithik Nandi, Amlan Sinha. **Committing Security of BBB Secure MACs.**
[ePrint]
3. Avik Chakraborti, Bishwajit Chakraborty, Nilanjan Datta, Avijit Dutta, Ashwin Jha, Sougata Mandal, Hrithik Nandi, Mridul Nandi, Abishanka Saha. **Naor-Reingold goes Beyond-the-Birthday-Bound.**
[ePrint]

INVITED AND CONTRIBUTED TALKS

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| Mar. 2026 | HCTR+: An Optimally Secure TBC-based Accordion Mode
FSE 2026, Singapore |
| Mar. 2026 | Short-Input Random Oracle from Public Random Permutations
ASK 2026, NTU, Singapore |
| Dec. 2025 | Sequential Indifferentiability of STH and EDM
Crypto Winter School 2025, IIT Bhilai, India |
| Dec. 2024 | Optimally Secure TBC Based Accordion Mode
ASK 2024, TCG CREST, Kolkata, India |
| Sep. 2024 | Efficient Variants of TNT with BBB Security
ProvSec 2024, Griffith University, Gold Coast, Australia |

WORKSHOPS AND CONFERENCES ATTENDED

- | | |
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| Mar. 23 – 27, 2026 | Fast Software Encryption (FSE 2026)
Singapore, Singapore |
| Mar. 19 – 22, 2026 | Asian-workshop on Symmetric Key Cryptography (ASK 2026)
NTU, Singapore |
| Sep. 01 – 05, 2025 | Workshop on Generic Attacks and Proofs in Symmetric Cryptography
(GAPS 2025), NTU, Singapore |
| Dec. 14 – 17, 2024 | Asian-workshop on Symmetric Key Cryptography (ASK 2024)
Kolkata, India |
| Dec. 09 – 13, 2024 | Asiacrypt 2024
Kolkata, India |

Sep. 25 – 27, 2024	International Conference on Provable and Practical Security (ProvSec 2024), Gold Coast, Australia
Dec. 10 – 13, 2023	Indocrypt 2023 Goa, India
Dec. 11 – 14, 2022	Indocrypt 2022 Kolkata, India

ACADEMIC SERVICES

Sub-reviewer | CRYPTO 2026

Teaching Assistant	Discrete Mathematics, IAI TCG CREST, Aug. 2024 – Dec. 2024 Cryptology, IAI TCG CREST, Aug. 2023 – Dec. 2023
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REFERENCES

Dr. Avijit Dutta
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Dr. Nilanjan Datta
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Institute for Advancing Intelligence, TCG CREST, Kolkata, India
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DECLARATION

I hereby declare that the information stated above is true to the best of my knowledge and belief.

Place: Kolkata, India

Hrithik Nandi